Towards Improving Deep Embedded Document Clustering with Long Range Text Semantics & Neighborhood Documents Information

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1 INTRODUCTION & MOTIVATION

Document clustering is a fundamental task in Natural Language Processing (**NLP**) that aims to group semantically similar documents without labeled supervision. Traditional clustering methods based on lexical representations often struggle to capture contextual semantics and complex neighborhood relationships in large scale text corpora.

Recent deep clustering approaches combine:

1. Semantic representation learning
2. Latent embedding optimization
3. Graph-based structural modeling

However, existing graph-based clustering frameworks still suffer from:

1. Static graph dependency
2. Semantic-structural inconsistency
3. Graph sensitivity
4. Scalability limitations
5. Unstable optimization behavior

This research investigates adaptive semantic structural learning strategies for improving deep embedded document clustering.

2 RESEARCH OBJECTIVES

The thesis proposes two major objectives:

1. Obtain the long-range semantic relationships using a Transformer guided semantic representation refinement, thus improving the representation of the content of documents.
2. Use neighborhood structural information via adaptive graphs construction and GCN modelling for the refinement of clustering while training.

3 BASELINE FRAMEWORK

Document Embeddings:

1. **TF-IDF** lexical representations
2. **SBERT** contextual semantic embeddings

Clustering Models

1. **KMeans**
2. **IDEC**
3. **SDCN**

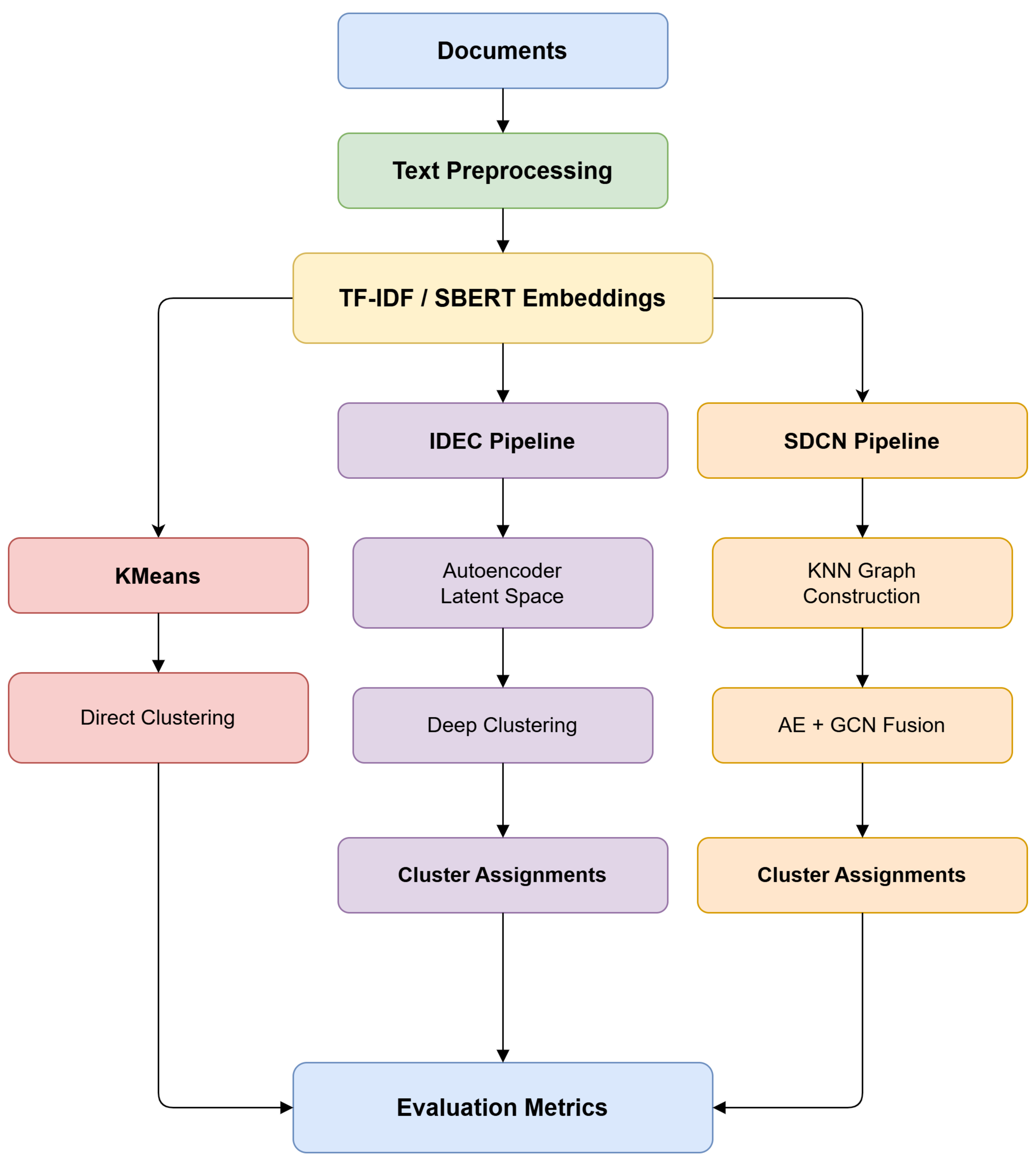
Graph Construction

1. Cosine similarity **KNN** graphs
2. Symmetric graph normalization
3. Sparse adjacency propagation

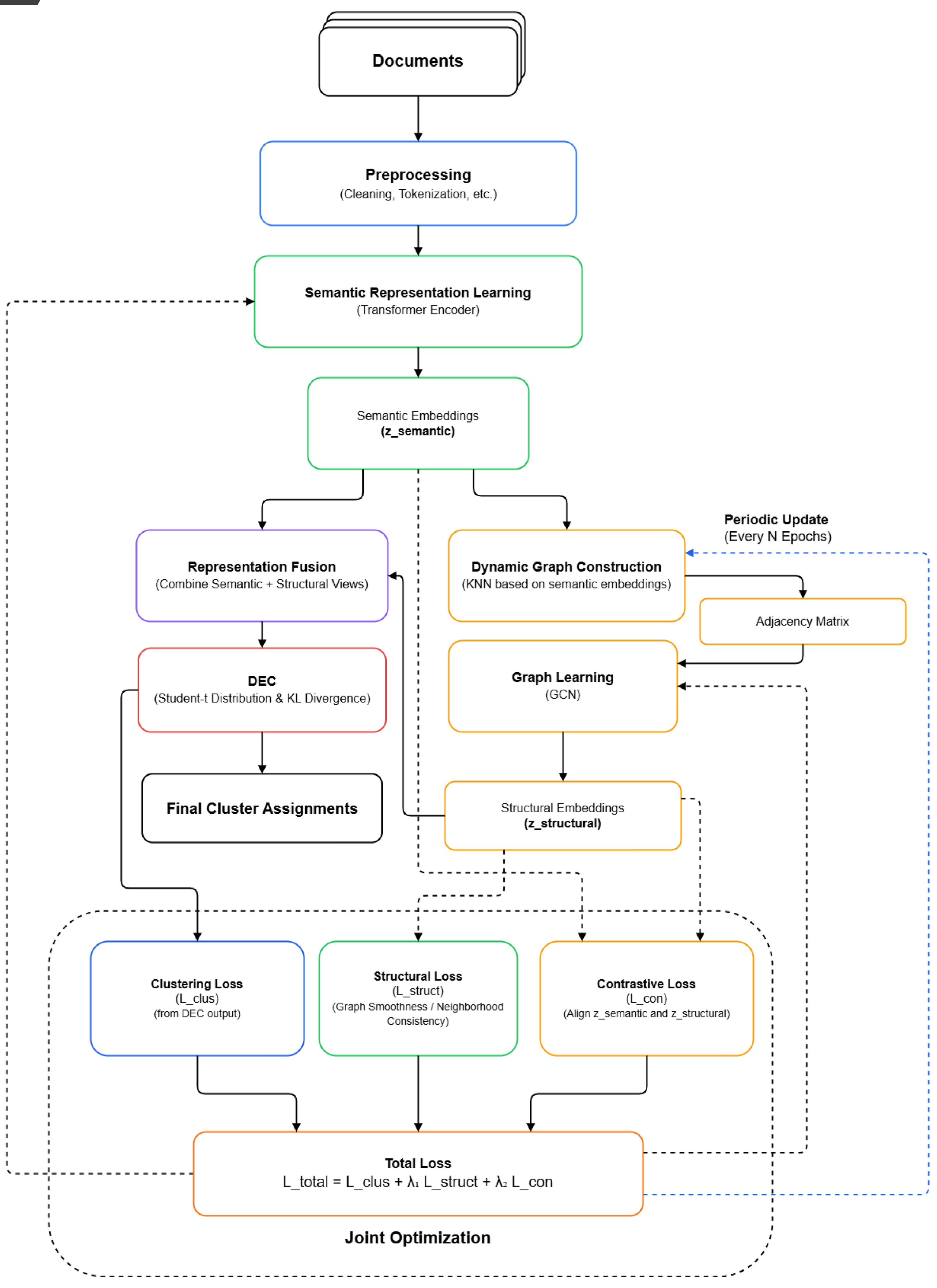
Evaluation Metrics

1. Accuracy (**ACC**)
2. Normalized Mutual Information (**NMI**)
3. Adjusted Rand Index (**ARI**)
4. Cluster Purity

4 BASELINE PIPELINE



5 PROPOSED MODEL



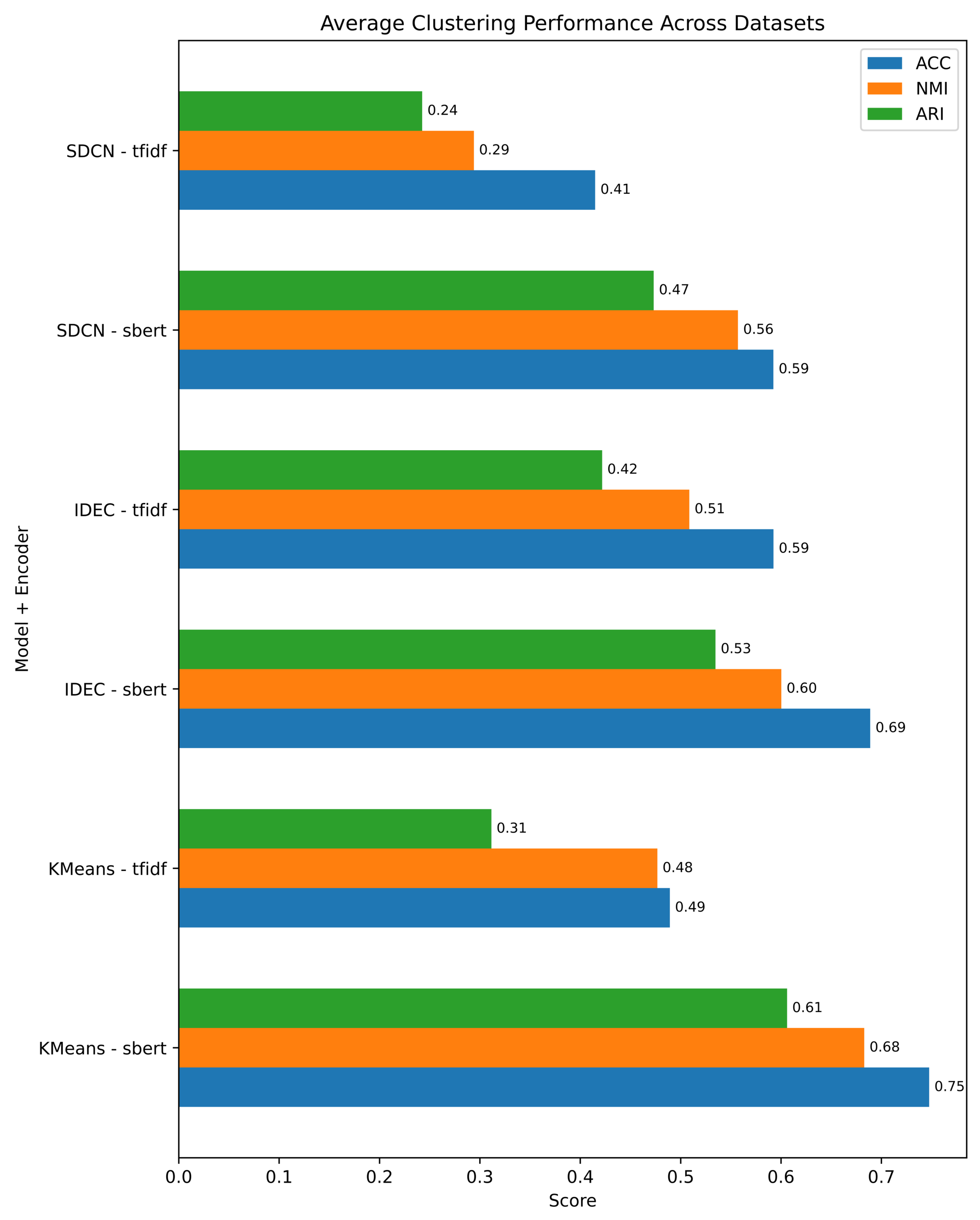
6 KEY FINDINGS

1. Contextual semantic embeddings significantly improve clustering quality compared to lexical representations.
2. KMeans operating directly on SBERT embeddings achieved highly competitive clustering performance.
3. Deep clustering architectures occasionally degraded pretrained semantic structure during latent compression.
4. Graph-based clustering methods were highly sensitive to graph construction quality and neighborhood structure.

7 EXPERIMENTAL SCORES

Encoder	ACC	NMI	ARI	Purity
SBERT	0.6763	0.6135	0.538	0.7482
TF-IDF	0.4989	0.4265	0.3253	0.5797

Model	ACC	NMI	ARI	Purity
IDEC	0.6407	0.5545	0.4783	0.7277
KMeans	0.6184	0.5799	0.4587	0.6955
SDCN	0.5036	0.4256	0.3578	0.5686



8 FUTURE WORK

1. Dynamic graph reconstruction
2. Adaptive semantic refinement
3. Contrastive semantic-structural learning
4. Robustness & ablation analysis
5. Scalability evaluation on large corpora